

# Greek fortition in Logical Phonology\*

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Logical Phonology (henceforth, LP; e.g., Bale et al. 2014, Bale and Reiss 2018, Reiss 2021, Dabous et al. 2024, Gorman and Reiss 2024, Reiss 2024) is a formally rigorous, *substance-free* theory of phonology. LP posits that features are binary-valued (rather than privative) and segments are thus sets of valued features linked to a X-slot. A segment must be *consistent*: if a segment  $\varsigma$  is specified +F, it cannot also be specified –F or vice versa. However, a segment can be *underspecified*:  $\varsigma$  need not contain any specification for F at all. Following Poser (1982) et seq., the distinction between feature-filling and feature-changing rules is derived by decomposing feature-changing processes (e.g., voice assimilation) into separate, independently motivated feature insertion (*unification*) and deletion (*subtraction*) operations.

(1) Unification: If  $A$  and  $B$  are feature specifications, then  $A \sqcup B = A \cup \{cF \mid cF \in B \wedge -cF \notin A\}$ .

(2) Subtraction: If  $A$  and  $B$  are feature specifications, then  $A \setminus B = \{cF \mid cF \in A \wedge cF \notin B\}$ .

Unification provides a novel solution to the decades-old problem (e.g., Archangeli 1988, Bonet 1995, Lees 1961:12–14, Lightner 1971:236) of targeting relatively underspecified structures without also affecting more richly specified ones. It is also a principled way to implement Sharon Inkelas and colleagues’ (e.g., Inkelas and Cho 1993, Inkelas 1995, Inkelas and Orgun 1995, Inkelas et al. 1997) notion of *inalterability as prespecification*, according to which segments underspecified for some feature  $F$  are *mutable* with respect to processes inserting  $\pm F$ , allowing many patterns which were previously analyzed using *morphophonological* rules, morpheme-specific constraints, cophonologies, or lexical diacritic features to be generated by the narrow phonology.

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Markopoulos and Revithiadou (2023), henceforth M&R, describe a process of manner dissimilation in Modern Greek by which certain labial fricatives, those usually spelled ⟨φ, β⟩ as in (3ab), strengthen—and devoice in the case of ⟨β⟩—to [p] before the sibilant /s/; they are realized as [f, v] elsewhere. However, some labial fricatives, usually spelled ⟨υ⟩ as in (3c), resist fortition in this context.<sup>1</sup>

(3) Fortition (M&R; additional data from Despina Nifakos, p.c.):

|    | IPFV.1SG /-o/ | PST-1SG /-s-a/        |             |
|----|---------------|-----------------------|-------------|
| a. | [ˈɣrafo]      | [ˈɣrap <b>sa</b> ]    | ‘write’     |
|    | [ˈvafo]       | [ˈevap <b>sa</b> ]    | ‘paint’     |
|    | [ˈstefo]      | [ˈestep <b>sa</b> ]   | ‘crown’     |
| b. | [ˈravo]       | [ˈrap <b>sa</b> ]     | ‘sew’       |
|    | [ðuˈlevo]     | [ðule <b>psa</b> ]    | ‘work’      |
|    | [peˈðevo]     | [peðep <b>sa</b> ]    | ‘pester’    |
| c. | [vraˈvevo]    | [vraˈvef <b>sa</b> ]  | ‘award’     |
|    | [ermiˈnevo]   | [ermiˈnef <b>sa</b> ] | ‘interpret’ |
|    | [peˈðevo]     | [peˈðef <b>sa</b> ]   | ‘train’     |

Note especially the homonym ⟨παιδεύω⟩ [peˈðevo], which means ‘pester’ if it undergoes fortition and stress shift in the past tense to [peðep**sa**], and ‘train’ if it does not.

M&R cite a number of acoustic studies confirming phonetic equivalences and conduct a formal acceptability judgment task ( $n = 35$ ); they then analyze this pattern using Gradient Harmonic Grammar (Smolensky and Goldrick 2016). However, this data can also be generated using the more modest architectural assumptions of LP. Suppose that *inalterable* /v/ is underlyingly +CONTINUANT /v/, whereas *mutable* /F, V/ denote labial fricatives underspecified for this feature.

<sup>1</sup>Abbreviations used: IPFV: imperfective; PST: past, 1SG.: first-person singular. Some past-tense forms also show an prothetic [e] and/or stress-shift in the past tense.

(4) Feature specification (partial):

|            | /F/ | /V/ | /v/ |
|------------|-----|-----|-----|
| CONTINUANT |     |     | +   |
| VOICE      | −   | +   | +   |

(5), a unification rule, strengthens /F, V/ when followed by a sibilant, but applies vacuously to inalterable /v/, which is already specified for CONTINUANT. It is similarly vacuous when applying to other labial consonants, such as /p, b, m/. A later context-free redundancy rule specifies /F, V/ not targeted by (5) as fricatives, and similarly applies vacuously to fully-specified segments including other labials. For completeness, the voice assimilation rules—which are implemented as a subtraction followed by a unification—critical orderings, and sample derivations are also provided.

(5) Fortition:  $\left[ +\text{LABIAL} \right] \sqcup \left\{ -\text{CONTINUANT} \right\} / - \left[ +\text{STRIDENT} \right]$

(6) Default frication:  $\left[ +\text{LABIAL} \right] \sqcup \left\{ +\text{CONTINUANT} \right\}$

(7) Voice assimilation (part 1):  $\left[ -\text{SONORANT} \right] \setminus \left\{ +\text{VOICE} \right\} / - \left[ \begin{array}{c} -\text{VOICE} \\ -\text{SONORANT} \end{array} \right]$

(8) Voice assimilation (part 2):  $\left[ -\text{SONORANT} \right] \sqcup \left\{ -\text{VOICE} \right\} / - \left[ \begin{array}{c} -\text{VOICE} \\ -\text{SONORANT} \end{array} \right]$

(9) Critical orderings: (5)  $\ll$  (6); (7)  $\ll$  (8)

(10) Sample derivations (stress omitted):

|              |                |                 |               |                |          |                   |
|--------------|----------------|-----------------|---------------|----------------|----------|-------------------|
| UR:          | yraF-o         | yraF-s-a        | raV-o         | raV-s-a        | vravev-o | vravev-s-a        |
| Rule (5):    |                | yrap <b>s</b> a |               | rab <b>s</b> o |          |                   |
| Rule (6):    | yra <b>f</b> o |                 | ra <b>v</b> o |                |          |                   |
| Rules (7–8): |                |                 |               | rap <b>s</b> a |          | vrave <b>f</b> sa |
| SR:          | yrafo          | yrapsa          | ravo          | rapsa          | vravevo  | vravefsa          |

LP provides all necessary representations and operations needed to implement Greek’s “exceptional” fortition pattern, previously analyzed by M&R using the much richer architectural primitives of Gradient Harmonic Grammar.

## References

- Archangeli, Diana. 1988. *Underspecification in Yawelmani Phonology and Morphology*. Garland.  
Revised version of 1984 MIT dissertation.
- Bale, Alan, Maxime Papillon, and Charles Reiss. 2014. Targeting underspecified segments: a formal analysis of feature-changing and feature-filling rules. *Lingua* 148:240–253.
- Bale, Alan, and Charles Reiss. 2018. *Phonology: A Formal Introduction*. MIT Press.
- Bonet, Eulàlia. 1995. Feature structure of Romance clitics. *Natural Language & Linguistic Theory* 13:607–647.
- Dabbous, Rim, Marjorie Leduc, Charles Reiss, and David Ta-Chun Shen. 2024. Locality is epiphenomenal: Adjacency is opaqueness. In *Actes du congrès annuel de l'Association canadienne de linguistique 2024 / Proceedings of the 2024 annual conference of the Canadian Linguistic Association*.
- Gorman, Kyle, and Charles Reiss. 2024. Metaphony in Substance Free Logical Phonology. *Phonology* to appear.
- Inkelas, Sharon. 1995. The consequences of optimization for underspecification. In *Proceedings of the North East Linguistic Society* 25, 287–302.
- Inkelas, Sharon, and Young-Mee Yu Cho. 1993. Inalterability as prespecification. *Language* 69:529–574.
- Inkelas, Sharon, and Cemil Orhan Orgun. 1995. Level ordering and economy in the lexical phonology of Turkish. *Language* 71:763–793.
- Inkelas, Sharon, Orhan Orgun, and Cheryl Zoll. 1997. The implications of lexical exceptions for the nature of grammar. In *Derivations and Constraints in Phonology*, ed. Iggy Roca, 393–418. Oxford University Press.
- Lees, Robert B. 1961. *The Phonology of Modern Standard Turkish*. Indiana University Publications.
- Lightner, Theodore M. 1971. On Swadesh and Voegelin's 'A Problem in Phonological Alternation'. *International Journal of American Linguistics* 37:227–237.

- Markopoulos, Giorgos, and Anthi Revithiadou. 2023. F as in *fricative* or as in *fortis*? Or both? Paper presented at the GLOW 46 Workshop on Non-Automatic Alternations in Phonology.
- Poser, William. 1982. Phonological representation and action at-a-distance. In *The Structural of Phonological Representations*, ed. Harry van der Hulst and Norval Smith, 121–58. Foris.
- Reiss, Charles. 2021. Towards a complete Logical Phonology model of intrasegmental changes. *Glossa* 6:107.
- Reiss, Charles. 2024. Specificity in rule targets and triggers: Two v's in Hungarian. In *Actes du congrès annuel de l'Association canadienne de linguistique 2024 / Proceedings of the 2024 annual conference of the Canadian Linguistic Association*.
- Smolensky, Paul, and Matt Goldrick. 2016. Gradient symbolic representations in grammar: The case of French liaison. Ms. ROA-1286. URL: <https://roa.rutgers.edu/article/view/1552.html>.